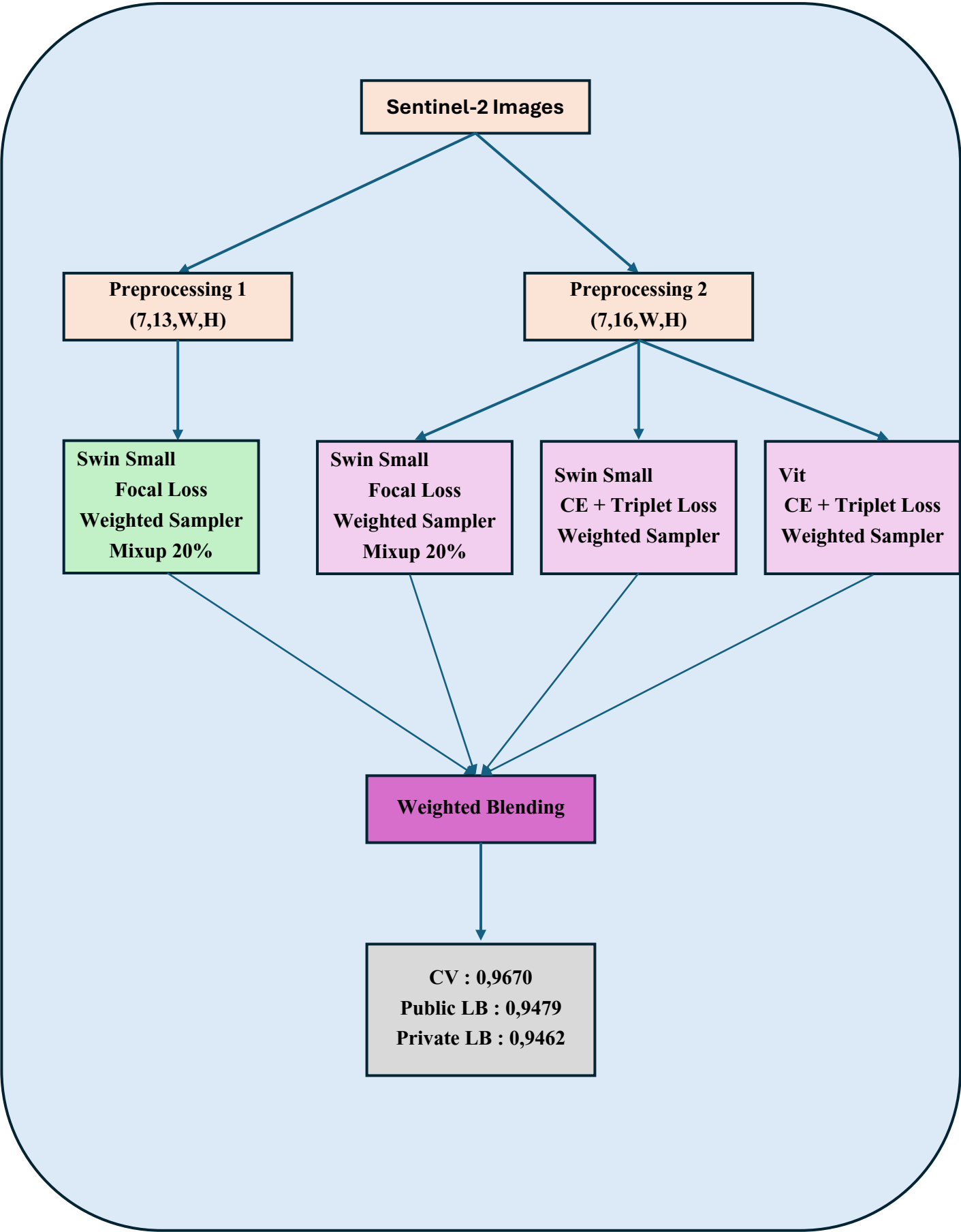


Spatio-Temporal Crop Classification Solution

Overview

Our solution employs a **spatio-temporal deep learning approach** tailored to Sentinel-2 imagery for crop classification (Rubber, Palm, Cocoa). By harnessing both spatial and temporal patterns across monthly images, we effectively modeled crop phenology and reduced label confusion.



Data Preparation

For each observation (ID), we stack the first 7 months of imagery, producing a tensor of shape (7, C, H, W) where C is the number of channels.

Channels:

- 13 channels: Standard approach using RGB (B4/B3/B2) + 10 key vegetation indices (NDVI, EVI, GNDVI, SAVI, NDWI, etc.).
- 16 channels: Extended by including raw Sentinel-2 bands B8 (NIR), B11, B12 in addition to the 13 above.
- Vegetation Indices: Computed per month using normalized bands, providing robust, crop-sensitive features

Data Split and Cross-Validation Strategy

- ❑ In order to maximize model generalization and minimize “location leak” between train and test, we designed a stratified spatial split for cross-validation.

Spatial Clustering : We extracted centroid coordinates from the training geometries.

- ❑ Using DBSCAN clustering with haversine distance (10 km radius), we grouped spatially proximate polygons into clusters, mimicking the spatial structure found in real deployment.
- ❑ Stratification To ensure that each fold is balanced for both spatial region and crop class, we concatenated label: `cluster_class = cluster_id + class_label`.
- ❑ We then performed a Stratified K-Fold split (n=5), stratifying on this label. This guarantees that each fold contains a representative mix of locations and target crops, without leaking neighboring parcels across folds.

Modeling Approach

Our architecture is built around **Swin Transformer Small** and **ViT backbones**.

Multi-Head Attention Pooling:

- After extracting per-month embeddings with the backbone, we apply a custom multi-head attention pooling mechanism, enhanced with positional embeddings for months.

This allows the model to learn the order and progression of crop phenology.

Loss Functions:

- Focal Loss with class-specific alpha (giving higher weight to underrepresented or confused classes like Palm and Rubber).
 - Cross-Entropy Loss + Triplet Loss: Encourages the model to push apart features for confused classes (Rubber vs. Palm).
 - Optimizer and Scheduler: Varied between models for diversity in ensembling.
- Sampling Strategy Weighted Random Sampler: To address class imbalance, we used a weighted sampler in training, which boosted overall F1 by +2-3%.

Ensembling

Our final submission blends the outputs of four models:

- ❖ Swin Small — Focal Loss, 13 channels (RGB + 10 VIs) [4h Run Time]
- ❖ Swin Small — Focal Loss, 16 channels [5h Run Time]
- ❖ Swin Small — CE + Triplet Loss, different optimizer & scheduler, 16 channels [5h Run Time]
- ❖ ViT — CE + Triplet Loss, different optimizer & scheduler, 16 channels [9h Run Time]